

RED OAK, IA, USA

WMO: 725494

Lat: 41.010N

Lon: 95.260W

Elev: 318

StdP: 97.56

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04966

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.8	-16.9	-23.9	0.5	-18.6	-21.3	0.6	-15.1	12.3	-4.0	11.3	-4.0	2.1	0	0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	34.9	25.4	33.0	24.7	32.2	24.4	27.4	32.7	26.4	31.6	25.3	30.3	5.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	22.4	31.1	24.9	20.8	30.0	23.8	19.3	28.9	89.3	32.6	83.8	31.2	79.7	30.2	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.2	9.7	8.6	DB	-25.0	36.6	3.5	1.6	-27.6	37.8	-29.6	38.7	-31.6	39.6	-34.1	40.7	
			WB	-25.0	28.5	3.7	1.1	-27.7	29.3	-29.9	30.0	-32.0	30.6	-34.7	31.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	-3.8	-1.7	4.9	11.4	17.5	22.8	24.8	23.4	19.1	12.3	4.9	-1.6
	DBStd	11.37	6.75	6.92	6.62	5.55	4.75	3.53	3.19	3.25	4.68	5.28	5.90	6.05
	HDD10.0	1565	429	328	185	47	3	0	0	0	1	36	173	362
	HDD18.3	3233	687	559	419	216	73	6	1	3	46	198	406	619
	CDD10.0	2013	0	2	26	90	236	385	458	414	276	106	19	1
	CDD18.3	642	0	0	2	8	48	141	201	159	71	11	2	0
	CDH23.3	6812	0	0	19	132	530	1472	2179	1549	766	132	33	0
	CDH26.7	2755	0	0	2	37	183	598	967	627	288	34	18	0
Wind	WSAvg	3.6	3.7	3.8	4.4	4.7	4.2	3.5	2.6	2.3	2.9	3.5	3.7	3.5
Precipitation	PrecAvg	907	23	27	55	94	135	141	116	99	76	67	44	26
	PrecMax	1335	48	62	118	218	258	343	297	228	185	220	115	72
	PrecMin	637	5	4	3	20	38	34	38	16	17	13	1	0
	PrecStd	189	11	16	35	44	61	83	62	58	43	50	29	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.8	18.9	26.0	31.2	34.1	35.1	37.4	36.2	34.9	30.2	27.0	17.0
		MCWB	7.7	12.2	15.6	18.4	21.7	24.6	26.4	25.7	23.6	20.5	18.0	12.7
	2%	DB	11.1	14.8	22.4	27.3	31.1	33.1	35.0	33.7	32.4	27.2	20.9	12.5
		MCWB	6.7	9.5	14.4	17.1	21.2	24.0	26.3	26.0	23.3	18.0	13.4	8.5
	5%	DB	7.6	11.1	18.8	24.0	28.0	32.2	33.0	32.2	30.0	23.9	17.2	9.8
		MCWB	4.5	7.1	12.7	15.2	19.5	23.4	25.6	25.7	22.6	16.9	11.6	6.8
	10%	DB	5.0	7.6	15.9	21.3	26.3	30.2	32.1	30.1	27.7	22.0	14.0	7.1
		MCWB	2.5	4.7	10.5	13.9	18.5	22.6	25.3	24.2	20.6	15.3	9.6	4.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.3	13.4	18.3	20.4	24.6	26.9	29.0	28.9	26.2	22.9	18.2	14.2
		MCDB	12.7	16.8	23.4	27.0	30.9	32.7	34.0	33.8	32.0	27.3	24.1	16.5
	2%	WB	6.7	9.9	15.8	18.5	22.6	25.6	27.6	27.3	24.7	20.1	14.9	9.4
		MCDB	10.4	14.6	20.1	24.9	28.4	31.2	32.7	32.3	30.2	24.0	18.9	11.5
	5%	WB	4.7	7.2	13.2	16.7	21.2	24.6	26.6	26.1	23.5	18.1	12.4	6.4
		MCDB	7.7	10.6	17.8	22.7	26.7	30.2	31.9	30.8	28.4	21.8	16.7	9.5
	10%	WB	2.7	4.9	10.5	15.0	19.7	23.5	25.6	24.9	22.1	16.0	9.9	4.1
		MCDB	5.0	7.6	15.4	20.1	24.7	29.1	30.6	29.3	26.5	20.5	14.0	6.7
Mean Daily Temperature Range	5% DB	MDBR	10.3	10.5	11.9	13.0	12.0	11.6	11.4	11.7	13.7	13.2	12.0	9.9
		MCDBR	14.6	15.0	17.4	17.6	14.8	13.0	12.7	13.3	15.0	16.4	16.9	14.2
		MCWBR	10.9	10.3	10.3	9.6	7.2	6.2	6.7	7.5	7.8	8.9	10.6	10.5
	5% WB	MCDBR	13.9	14.0	15.1	14.5	12.4	11.4	11.3	11.8	13.1	13.1	15.3	12.8
		MCWBR	10.6	10.2	9.7	9.0	6.8	6.6	7.0	7.4	7.4	8.4	10.6	9.9
Clear-Sky Solar Irradiance	taub	0.282	0.313	0.327	0.384	0.409	0.414	0.429	0.419	0.379	0.334	0.296	0.285	
	taud	2.450	2.379	2.405	2.265	2.264	2.300	2.248	2.292	2.359	2.472	2.508	2.455	
	Ebn at Noon	883	898	924	887	869	862	845	843	856	865	860	848	
	Edh at Noon	77	96	105	130	134	129	135	126	110	87	73	71	
All-Sky Solar Radiation	RadAvg	2.03	2.96	3.83	4.85	5.77	6.40	6.44	5.60	4.66	3.23	2.24	1.70	
	RadStd	0.17	0.19	0.35	0.33	0.46	0.52	0.45	0.34	0.31	0.38	0.17	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page